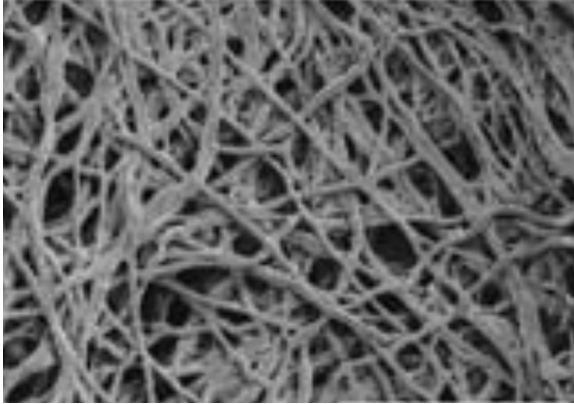
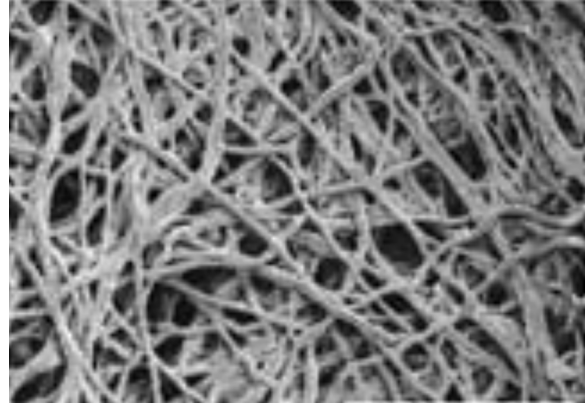


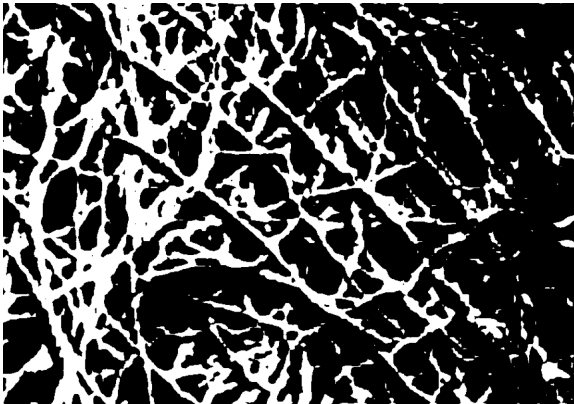
Original Image



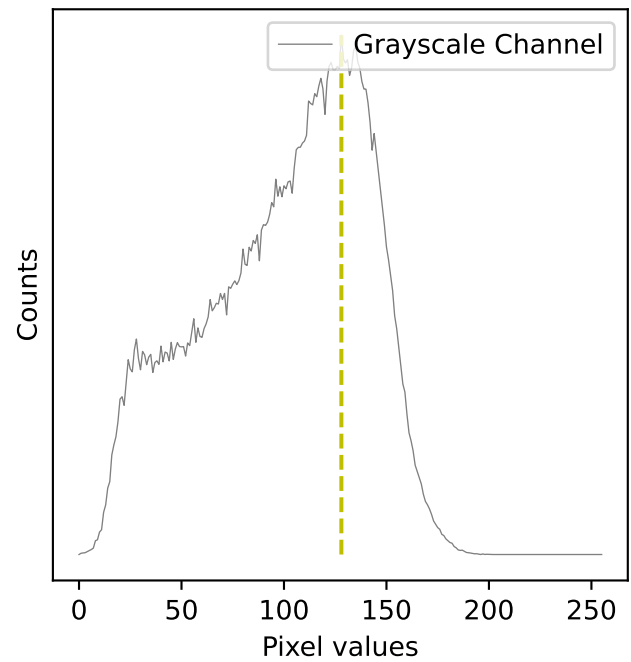
Grayscale Image



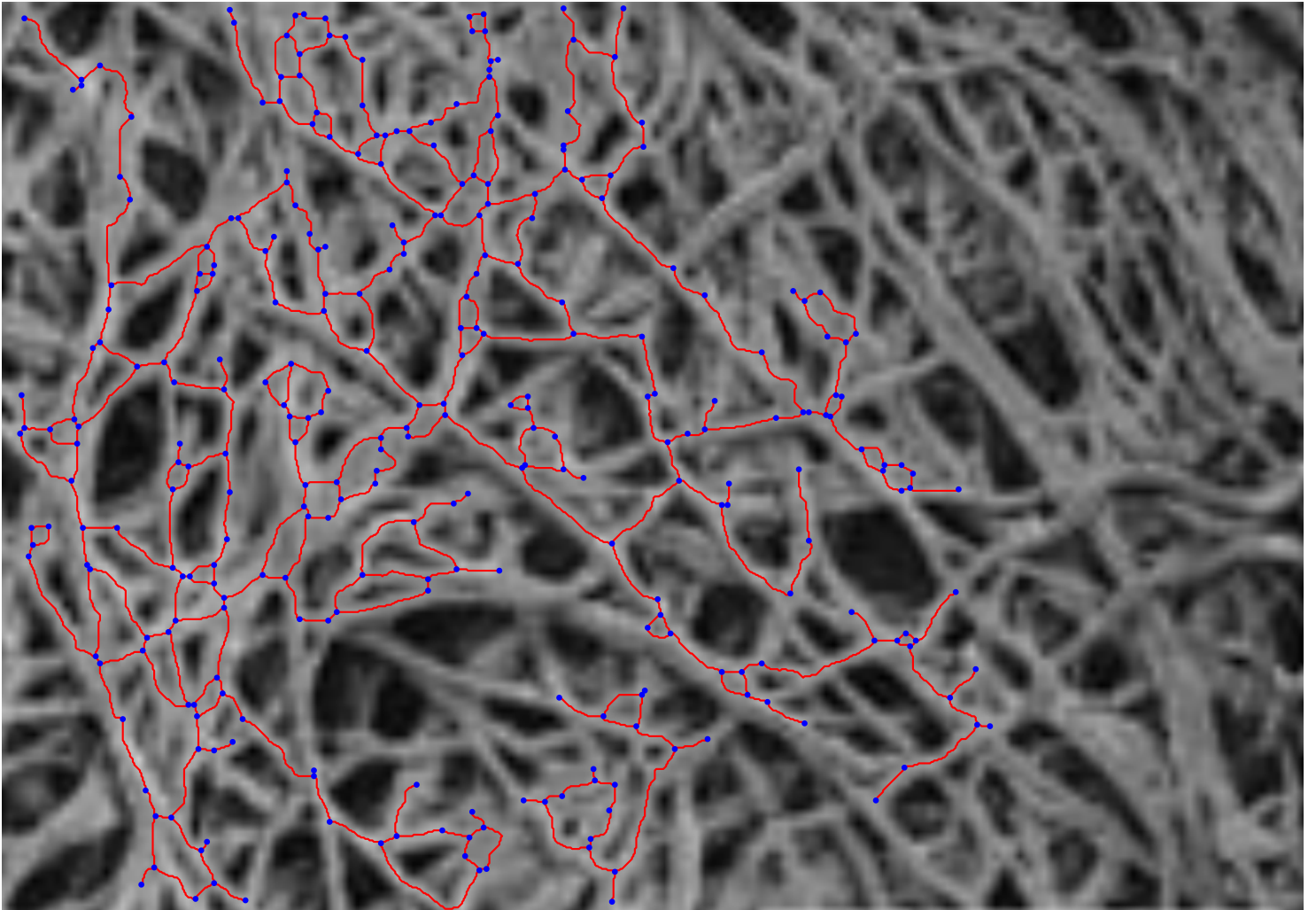
Binary Image



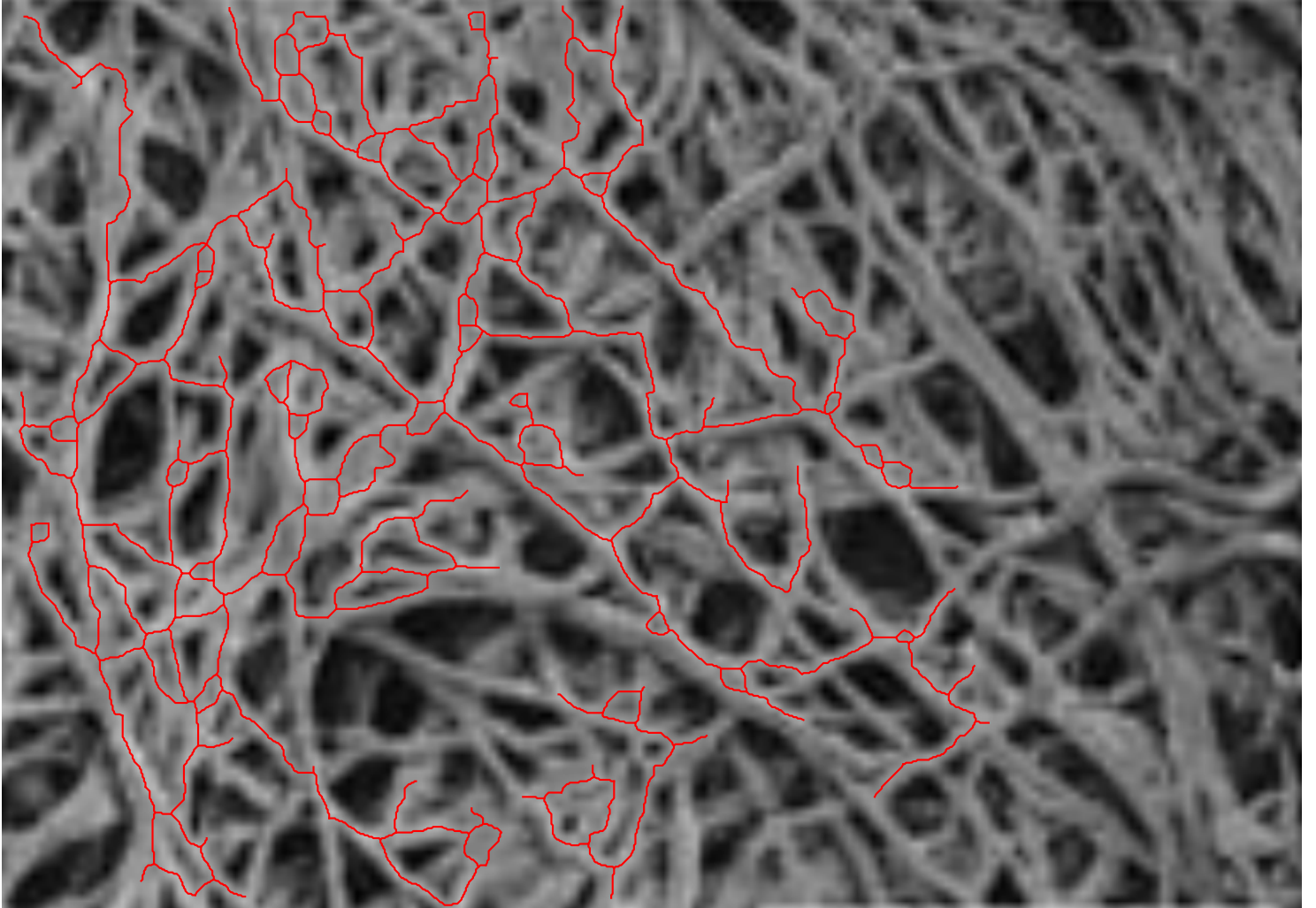
Histogram of Grayscale Image



Graph Node Plot



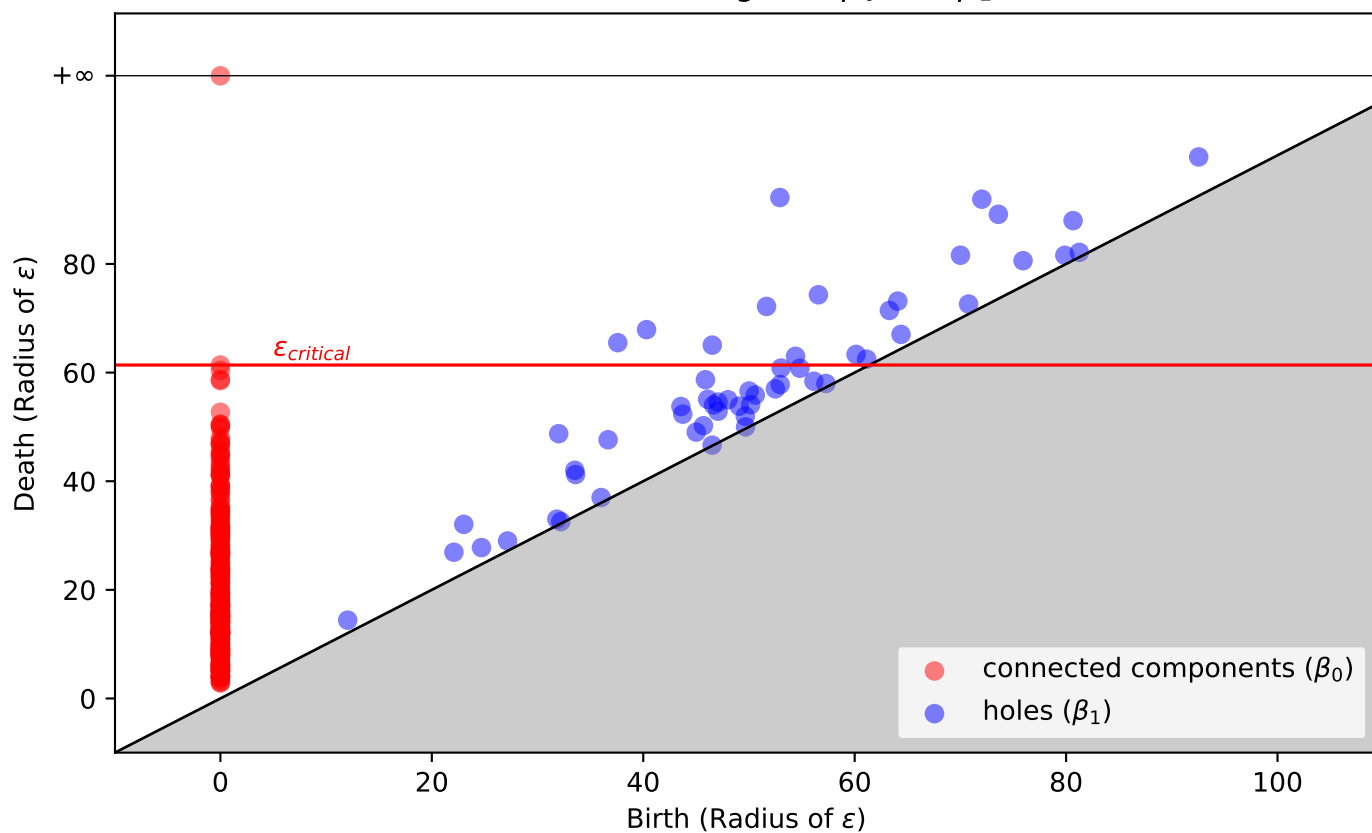
Graph Edge Plot



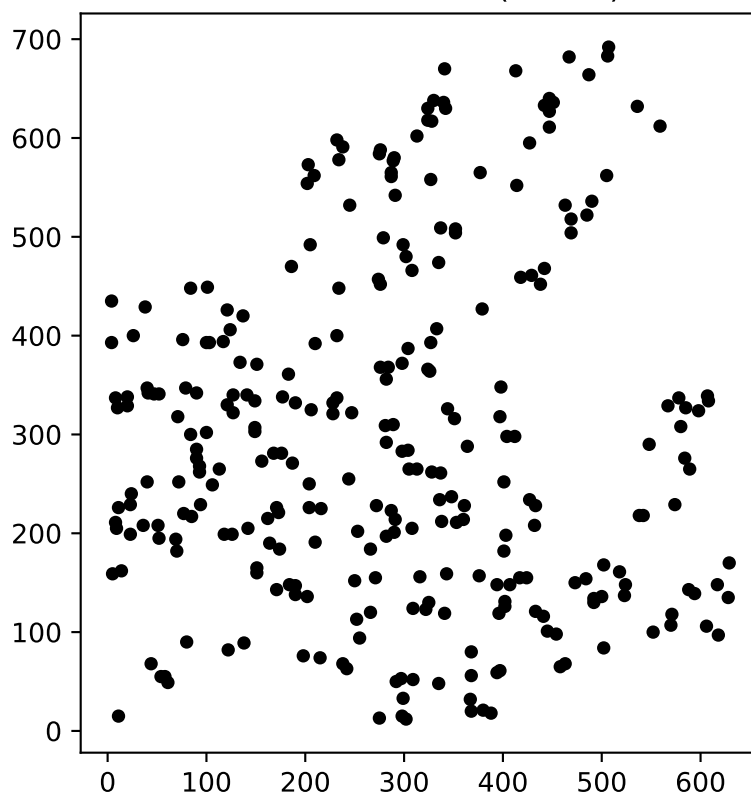
Unweighted GT parameters

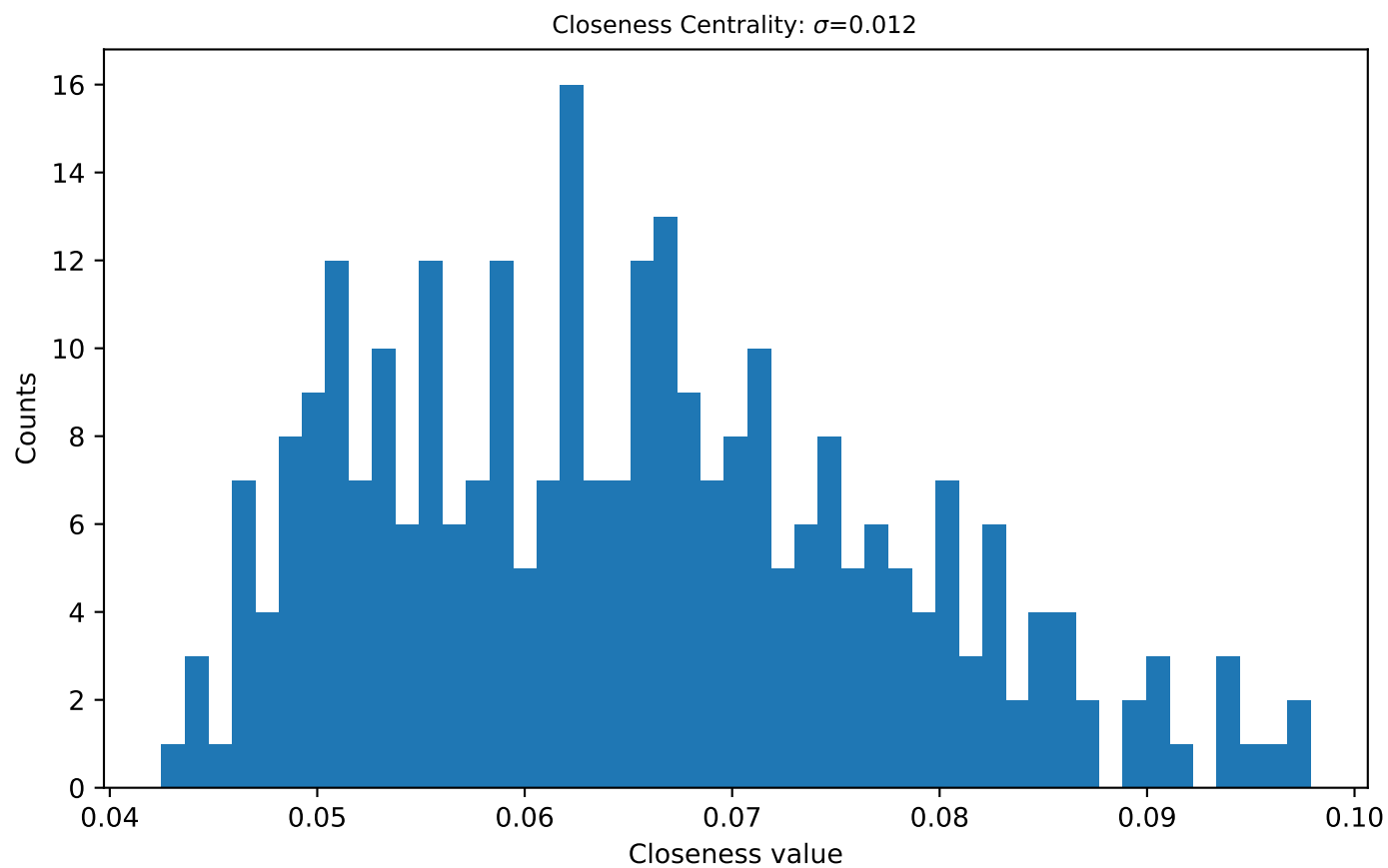
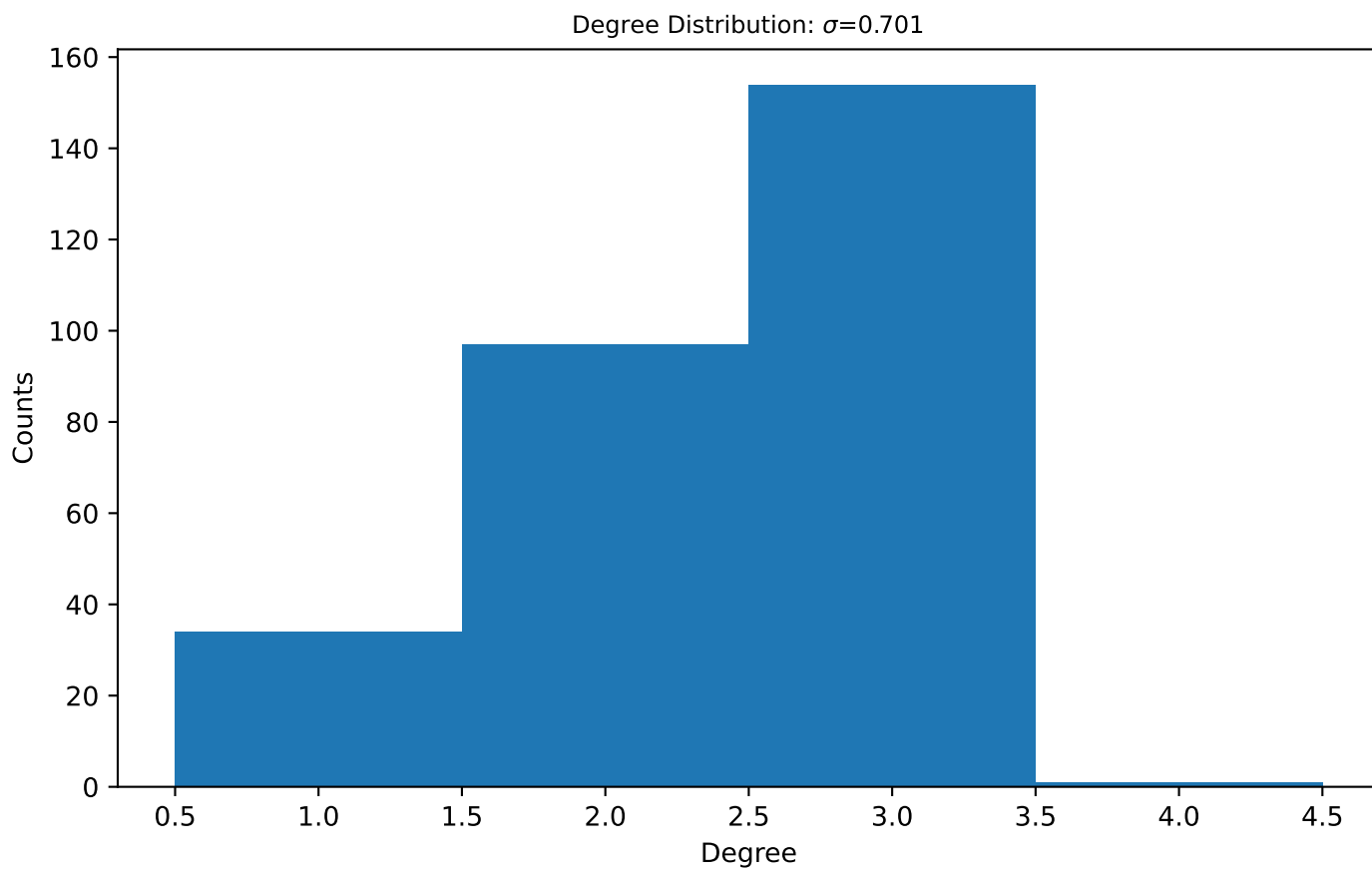
Number of nodes	286.0
Number of edges	347.0
Average edge angle (degrees)	164.153
Median edge angle (degrees)	87.138
Average degree	2.42657
Network diameter	35.0
Average node connectivity	1.32969
Graph density	0.00851
Global efficiency	0.09319
Wiener Index	646796.0
Assortativity coefficient	-0.02484
Average clustering coefficient	0.06294
Average betweenness centrality	0.05236
Average eigenvector centrality	0.02761
Average closeness centrality	0.06524
Persistent Homology--Critical Epsilon	61.39218
Persistent Homology--Active Holes Count	9.0

Persistence Diagram (β_0 and β_1)

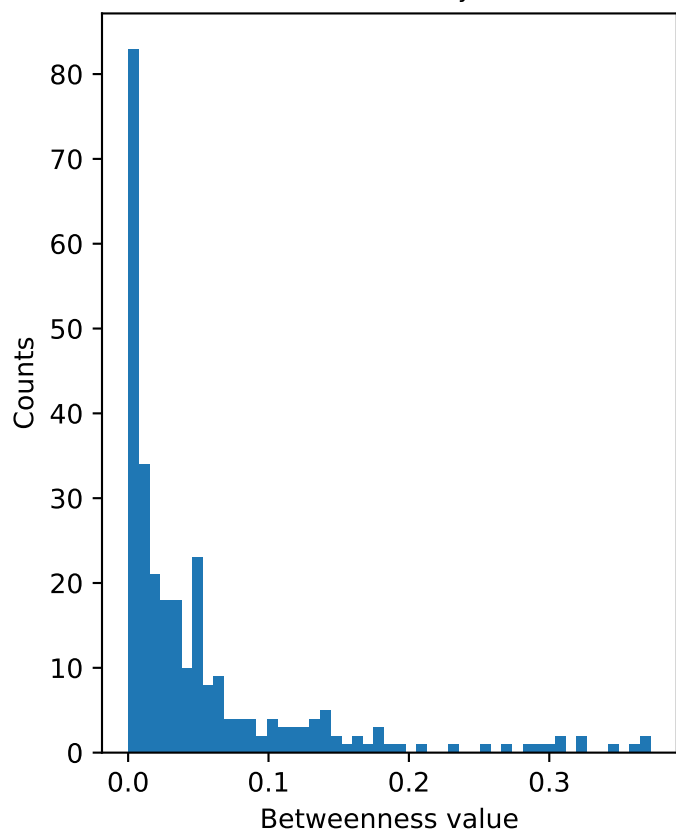


Point Cloud Data (Nodes)

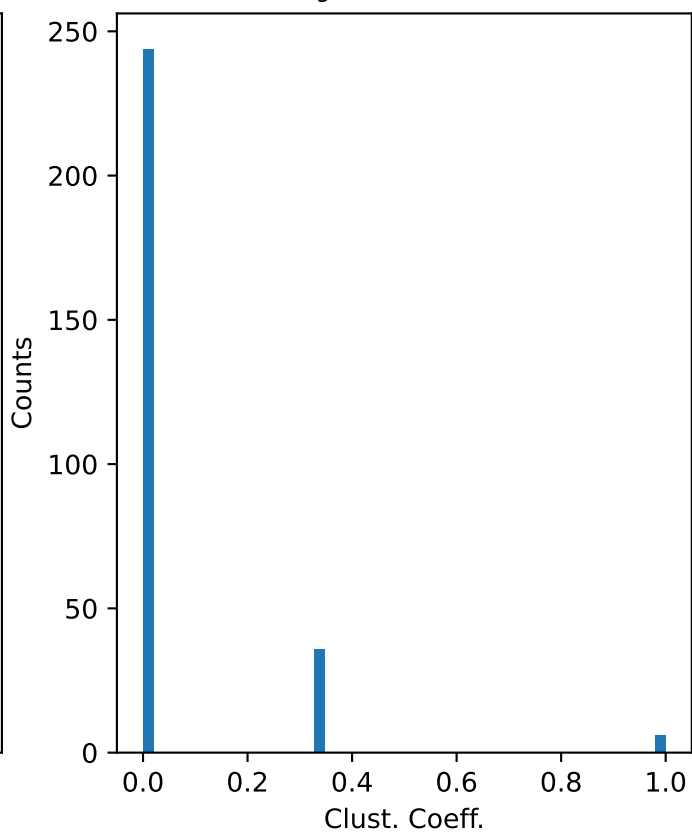




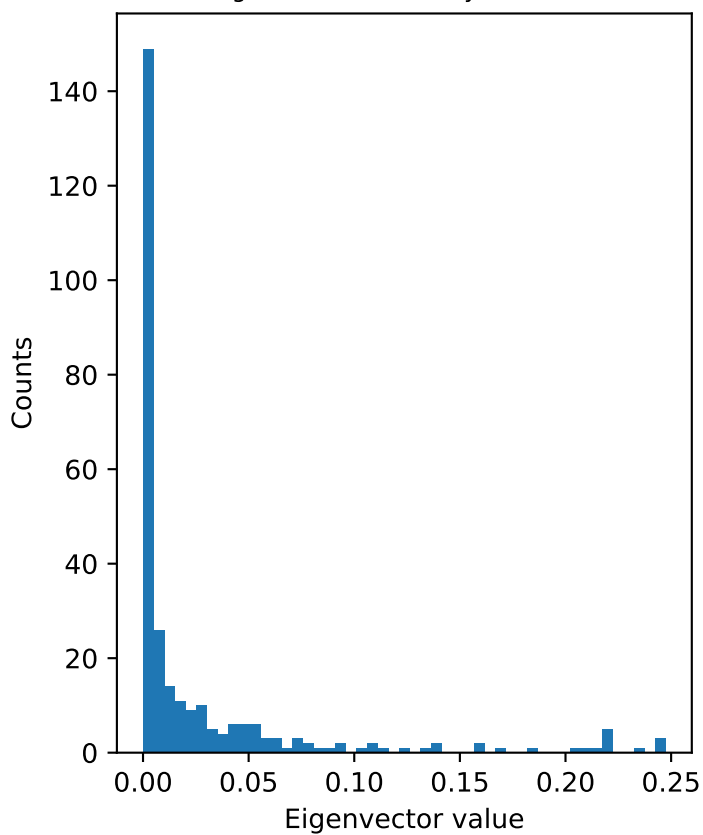
Betweenness Centrality: $\sigma=0.074$

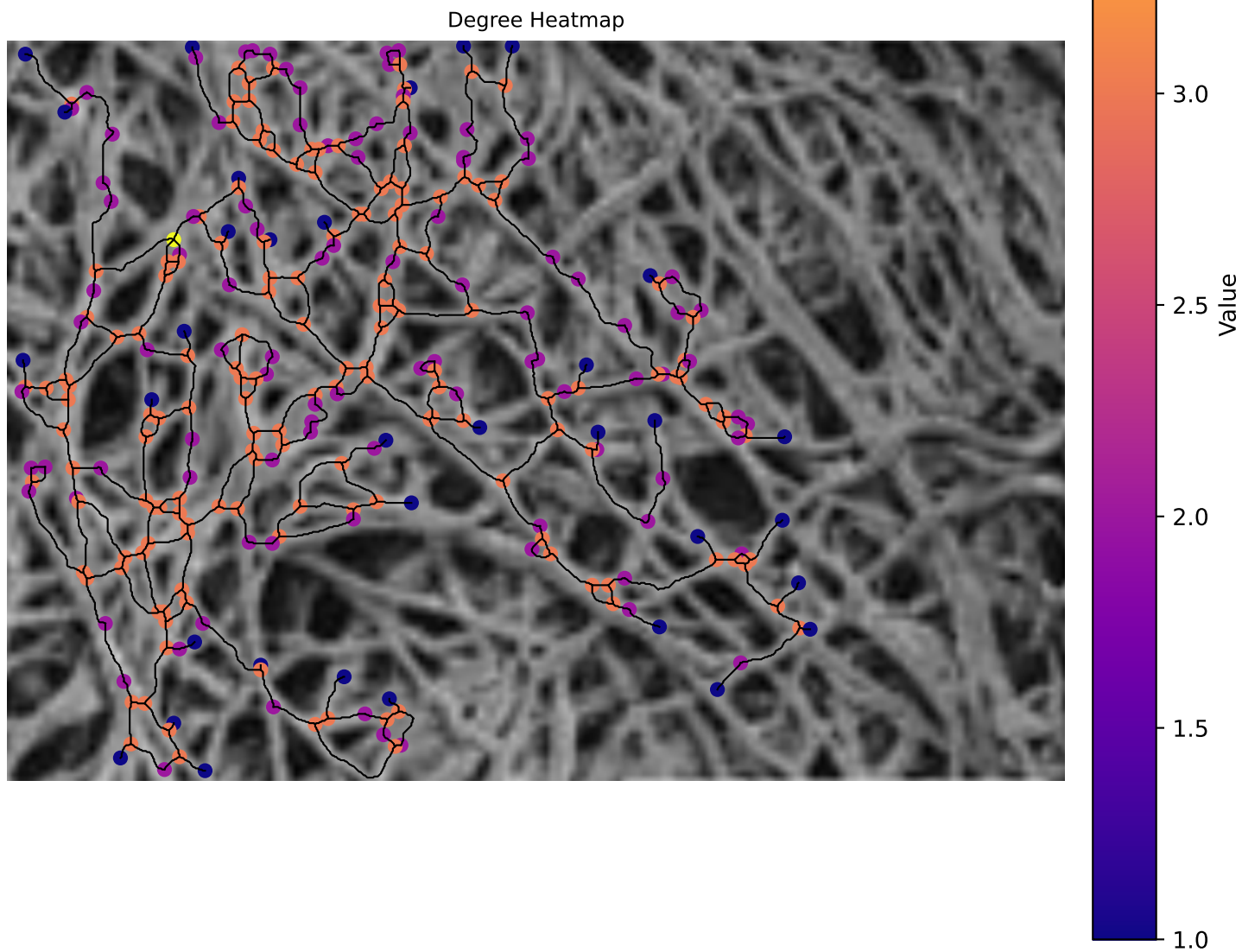


Clustering Coefficients: $\sigma=0.176$

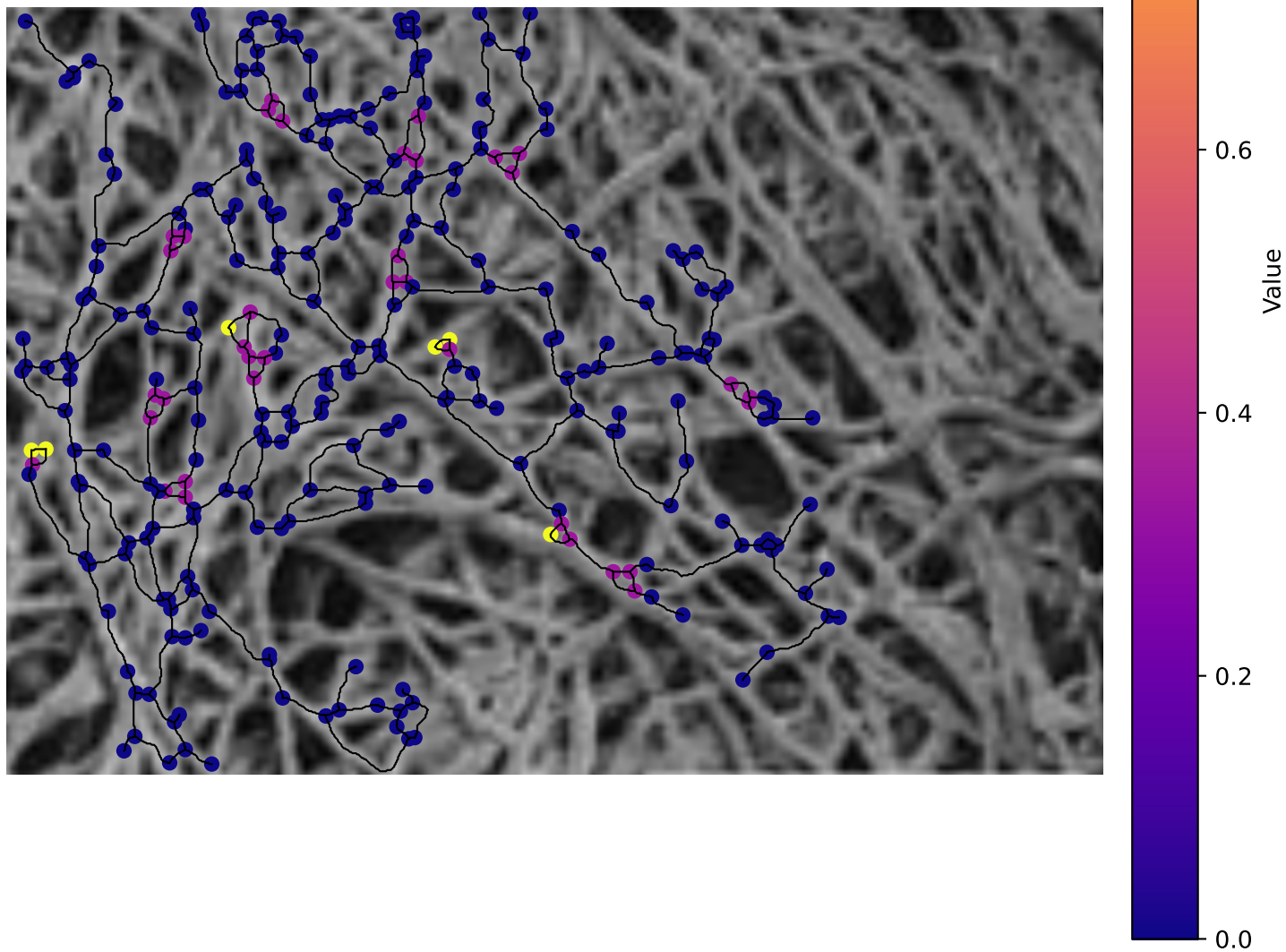


Eigenvector Centrality: $\sigma=0.052$

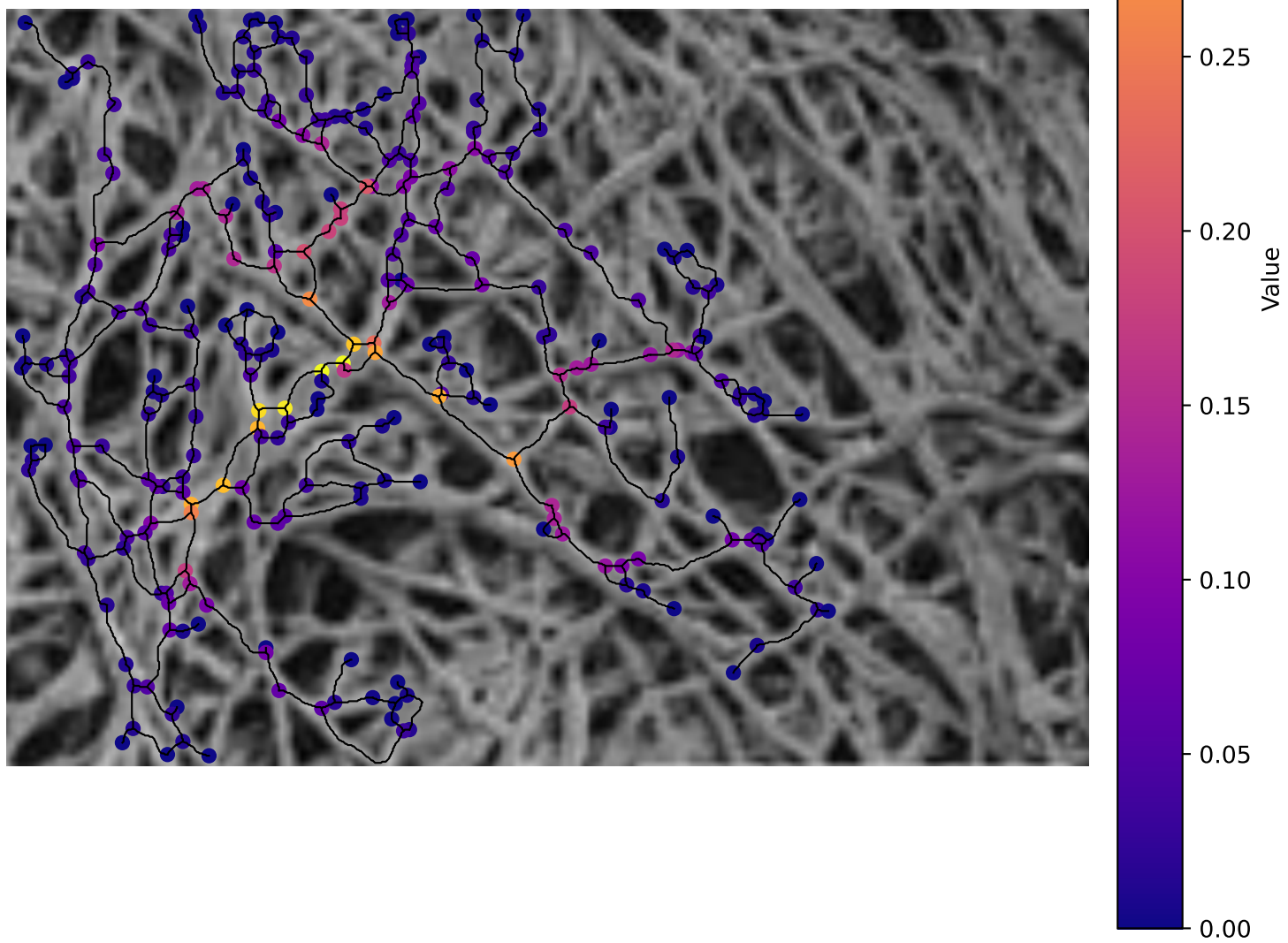




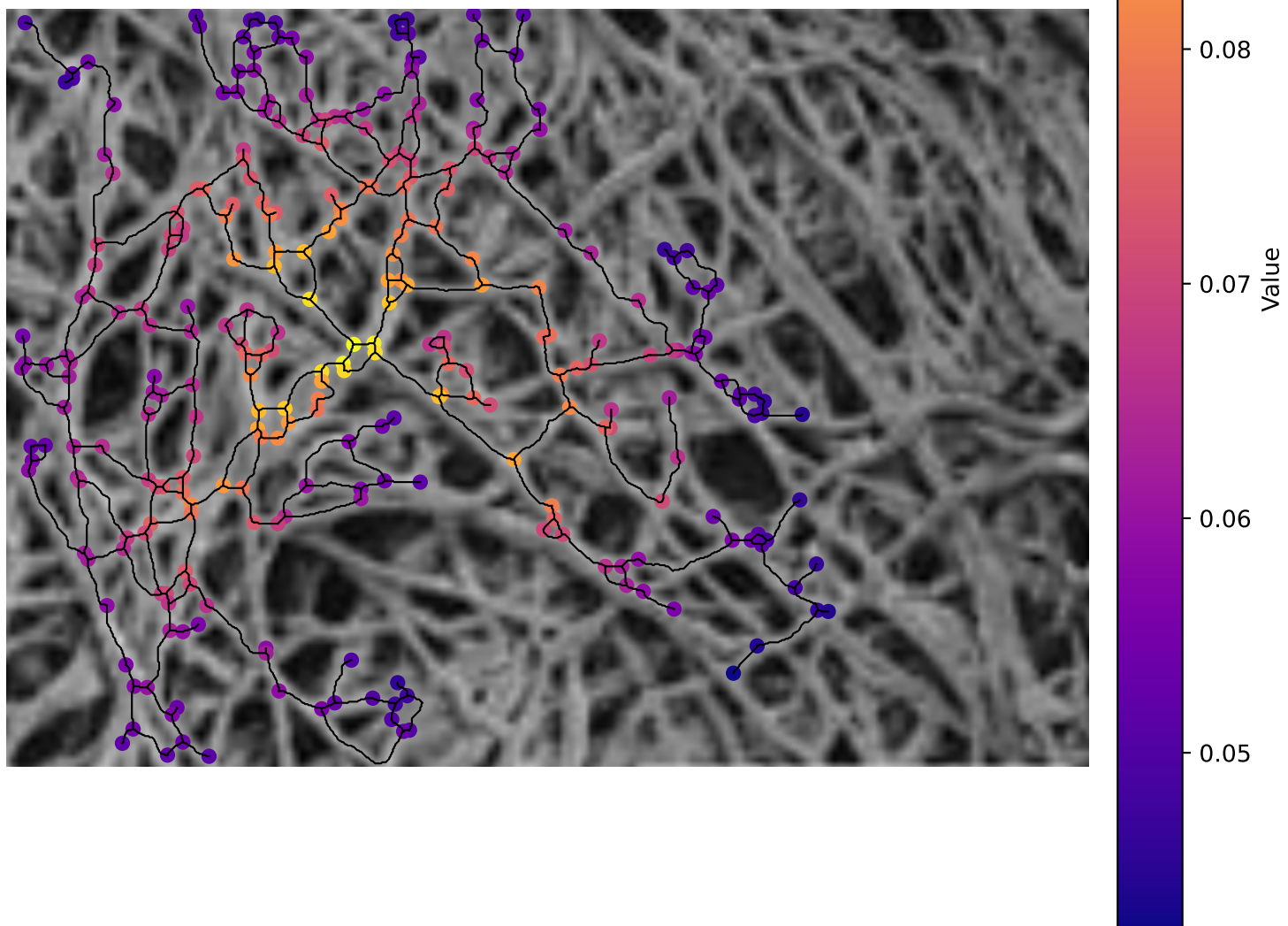
Clustering Coefficient Heatmap



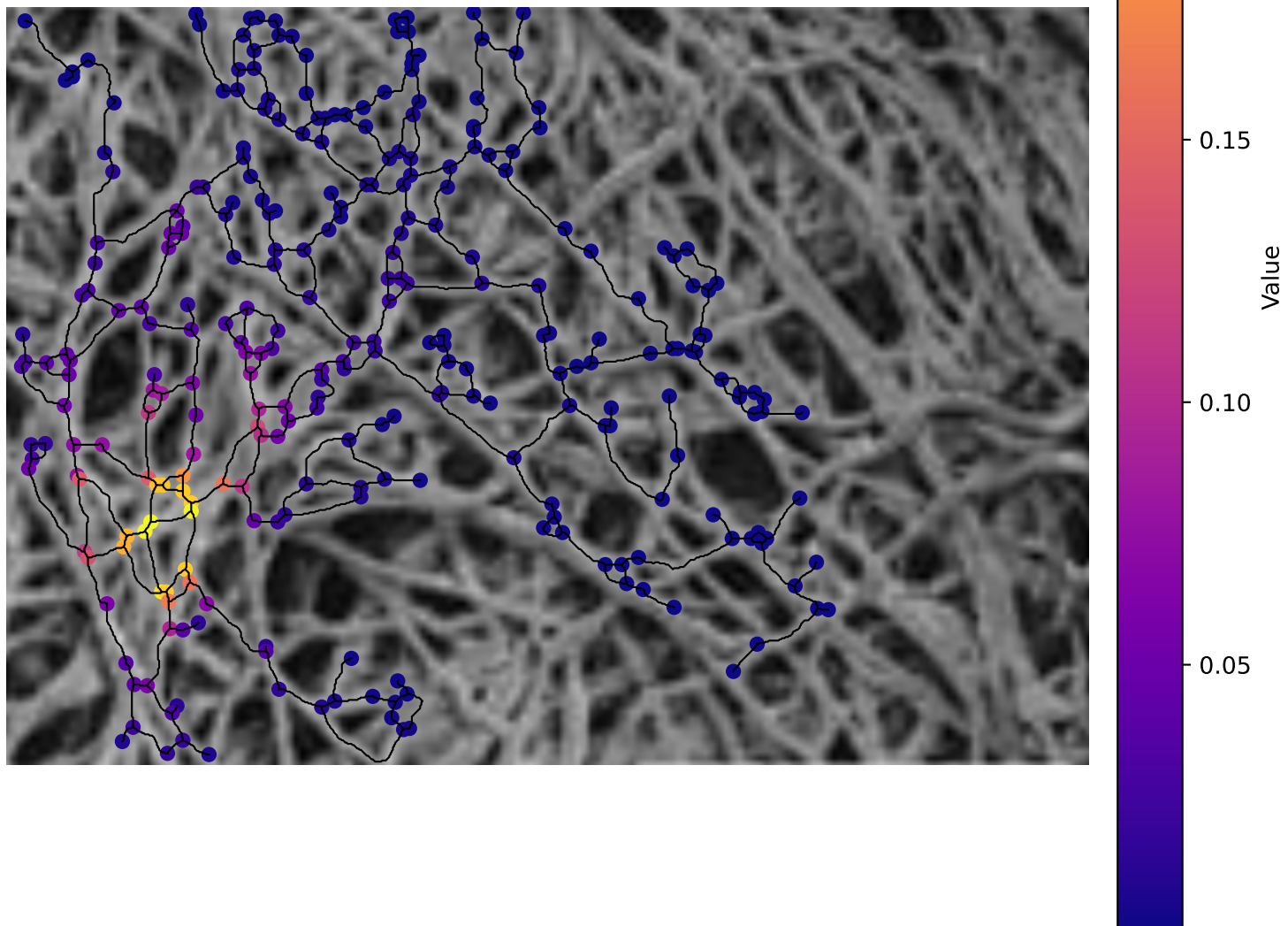
Betweenness Centrality Heatmap



Closeness Centrality Heatmap



Eigenvector Centrality Heatmap



Run Info

InVitroBioFilm.png
2026-06-03 17:31:51

Image Filter Configurations
Global Threshold (128.0) || Gamma = 1.0

Color Analysis
Eliminated Colors:

Microscopy Parameters
Scalebar Value = 0.0 nm || Scalebar Pixel Count = 1
Resistivity = 1.0Ωm

Image Scale
Size = 638 x 912 px || Scale Factor = 1.0

Graph Extraction Configurations
Merge Nodes || Prune Dangling Edges
Remove Objects of Size = 500 || Remove Self Loops